

3. Data Representation (Data Ka Roop)

Yeh chapter is baat ka base hai ki computer information ko kaise andar se dekhta hai. Computer ke liye har cheez— letters, photos, videos, games —sab kuch sirf **numbers** hain. Aur woh numbers hain **0 aur 1**. Toh inhi 0 aur 1 ko kis tarah manage kiya jaata hai, usi ka system "Data Representation" hai.

3.1 Measurement Units (Data Napne Ke Paemane)

Jaise hum doori ko meter mein, pani ko liter mein naapte hain, waise hi data ko napne ke liye kuch units hain. Sabse chhoti unit se shuru karte hain.

- **Bit (Binary Digit):** Data ki **sabse chhoti unit**. Iski value ya to **0** hogi ya **1**. Jaise paani ki ek boond, ya ek bulb jo ya to ON hai ya OFF.
 - *Analogy:* Ek simple switch. ON (1) ya OFF (0).
- **Nibble:** 4 bits ka group ek Nibble kehlata hai. Ye half byte hota hai.
 - **1010** = 1 Nibble.
- **Byte: 8 bits** ka group ek Byte kehlata hai. Ye computer memory ki **basic unit** hai. Ek byte mein ek character (A, B, C, ya 1, 2, 3) store ho sakta hai.
 - *Memory Trick:* "Ek **Byte** aapke naam ka ek **Letter** store kar sakta hai."
- **Word:** Ye CPU ke hisaab se change hota hai. Ek saath kitne bits process hote hain, use Word kehte hain. 16-bit, 32-bit, ya 64-bit word hota hai. 64-bit CPU ek saath 8 Bytes (64 bits) process kar sakta hai.

Ab aate hain badi units par. Yahan har level 1024 ka factor hai (kyunki ye 2 की power 10 hai), na ki 1000 ka. (Strictly memory ke liye, storage brands kabhi-kabhi 1000 use karte hain).

Unit	Full Form	Equal to (Memory)	Real-Life Analogy
1 KB	Kilobyte	1024 Bytes	Aadhe page ki simple text file.
1 MB	Megabyte	1024 KB	Ek high quality photo ya ek geet (MP3).
1 GB	Gigabyte	1024 MB	Ek HD movie (1.5-2 hrs) ya ek OS

			me.
1 TB	Terabyte	1024 GB	Modern laptop ki full hard disk. Hazon movies.
1 PB	Petabyte	1024 TB	Google, Facebook ke data centers ka scale. Laakhon movies.

Exam/Interview Tip: Kabhi puchein "How many bytes in 1 KB?" — jawab hai **1024**, 1000 nahi. (Purani storage devices par 1000 bhi maante the, lekin accurate 1024 hai).

3.2 Character Encoding Standards (Aksharo Ko Number Dena)

Computer sirf numbers jaanta hai. Toh **A** likhne ke liye usse ek number code chahiye. Isliye standards banaye gaye.

• ASCII (American Standard Code for Information Interchange):

- **Pronounced:** Ask-ee.
- **Bits:** 7-bit use karta hai (128 characters total). 8th bit error checking ke liye hoti thi.
- **Characters:** English ke upper/lower case letters (A-Z, a-z), numbers (0-9), punctuation marks (!, @, #...), aur control characters (Enter, Backspace).
- **Important Values Yaad Rakho:**
 - **'A'** = 65
 - **'a'** = 97
 - **'0'** = 48
 - **Space** = 32
- *Memory Trick:* "Amitabh Bachchan **65** saal ke hain. **a**am khao **97** din tak. **0** se shuru karo **48** baar."

• Unicode:

- **Problem:** ASCII mein Hindi, Chinese, Arabic, Japanese nahi likh sakte. Sirf English.
- **Solution:** Unicode aaya, jo duniya ki sabhi bhashaon ke har character ke liye unique code provide karta hai.
- **Bits:** 16-bit (65,536 chars), 32-bit etc.
- **UTF-8:** Unicode ka sabse flexible aur popular encoding. Ye 1 se 4 bytes tak use karta hai per character. English ke ASCII characters 1 byte mein

hi chale jaate hain, space bachata hai.

- *Real-World:* Aap apne phone mein ek saath Hindi, English, aur emoji 😊 type kar paate hain, ye Unicode ki wajah se hai.

✅ Topic 3: Summary & Revision

Key Point	Detail
Smallest Unit	Bit (0 or 1).
Basic Unit	Byte (8 Bits).
Nibble	4 Bits (Half Byte).
KB to PB	Multiply by 1024 at each step (KB -> MB -> GB -> TB -> PB).
ASCII	7-bit, 128 chars. 'A'=65, 'a'=97. English only.
Unicode	16/32 bit, universal, multi-language. UTF-8 most popular.

? MCQs

1. **1 Byte mein kitne bits hote hain?**

- A) 2
- B) 4
- C) 8
- D) 16

2. **1 MB mein kitne Kilobytes hote hain (technically)?**

- A) 1000
- B) 100
- C) 1024
- D) 10

3. **ASCII code of 'A' kya hai?**

- A) 97
- B) 65
- C) 48
- D) 32

4. **Duniva ki sabhi bhashaon ke live konsa encoding standard use hota**

... Dimaag ki kisi bhi cheez ko kehte hain. Kuch cheezon ka standard use hota hai?

- A) ASCII
- B) MORSE
- C) Decimal
- D) Unicode

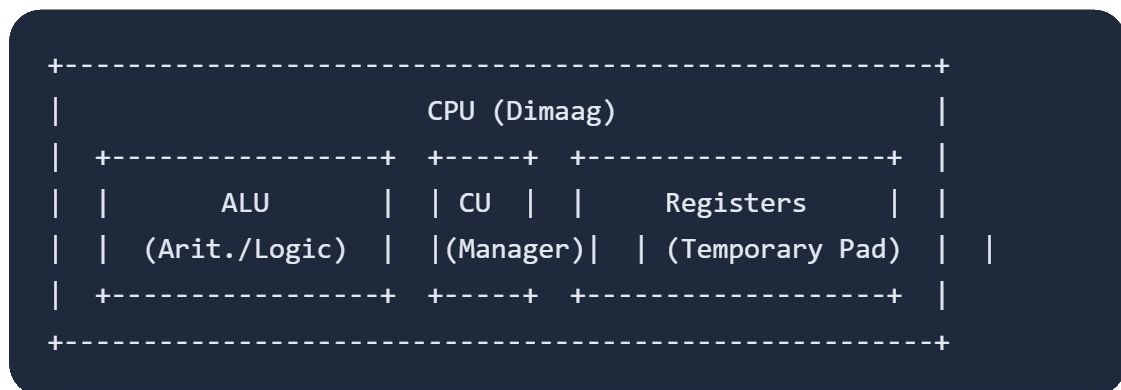
(Answers: 1-C, 2-C, 3-B, 4-D)

4. Computer Hardware (Computer ke Physical Parts)

Ab hum computer ko khol kar dekhenge. Jo parts aap chhoo sakte hain, physical cheezein hain, woh hardware kehlati hain.

Naya Term: CPU (Central Processing Unit) CPU computer ka **dimaag (Brain)** hai. Iska kaam hai instructions ko lena, samajhna, aur execute karna. Input aata hai, CPU process karta hai, Output deta hai.

CPU ke teen main parts hote hain:



1. ALU (Arithmetic Logic Unit):

- **Kaam:** Math aur logic. Jod, ghatana, guna, bhaag, aur AND, OR, NOT jaise faisle.
- *Analogy:* Ek calculator jo faisle bhi le sakta hai.

2. CU (Control Unit):

- **Kaam:** Computer ka **Manager**. Ye baaki sabhi parts ko direct aur coordinate karta hai. Data kahan se aana hai, kahan jaana hai, ALU ko kab kaam karna hai, memory se kab data lena hai. CU instructions fetch karta hai, decode karta hai, aur execute karwata hai.
- *Analogy:* Ek orchestra ka conductor. Khud koi instrument nahi bajata, lekin sabko signal deta hai kab bajana hai.

3. Registers:

• Registers:

- **Kaam:** CPU ke andar ki sabse tez, chhoti memory. Immediate data aur instructions store karne ke liye. ALU jo calculation kar raha hai, uske beech ka data yahin rakhta hai.
- *Analogy:* Chef jo kaam karte waqt apne saamne chhoti katori mein namak/masala rakhta hai. Wo uska register hai. Baar-baar bade dibbe (RAM/HDD) mein nahi jaata.

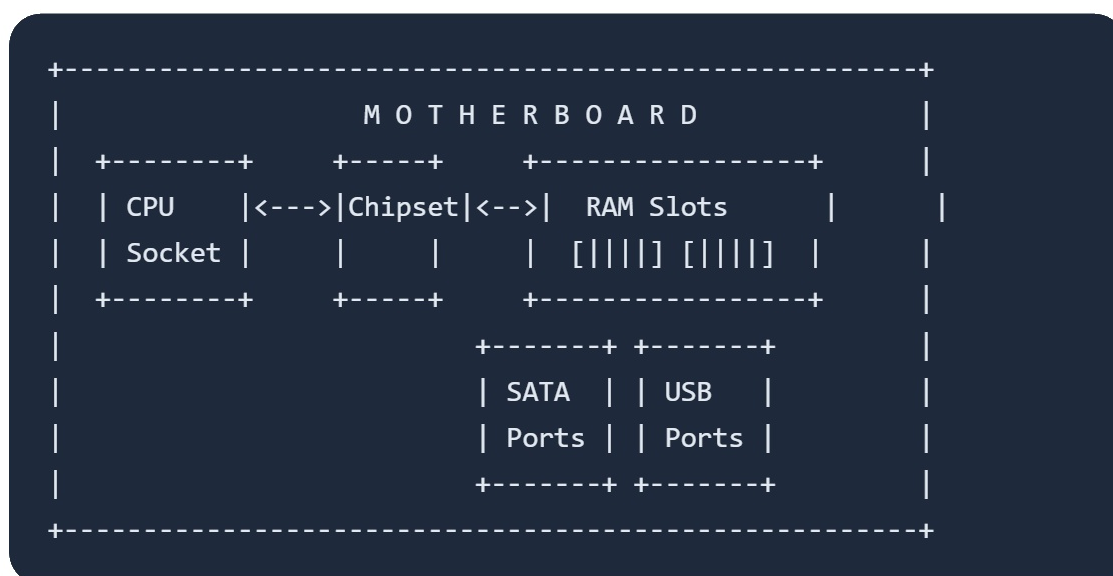
4.1 CPU ki Speed (Clock Speed)

CPU har kaam ek rhythm mein, ek clock ke signal par karta hai.

- **Clock Speed:** Ek second mein CPU kitne cycles complete kar sakta hai. Isse **Hertz (Hz)** mein naapte hain.
- **1 Hz** = 1 cycle per second.
- **1 MHz (Megahertz)** = 10,00,000 cycles per second. (10 Lakh)
- **1 GHz (Gigahertz)** = 1,00,00,00,000 cycles per second. (100 Crore)
- *Aaj kal ke CPU 3 GHz se 5 GHz tak ki speed se chalte hain.*
- *Analogy:* Jaise ek worker ghanti baje to ek piece assemble kare. Agar ghanti tez baje (higher clock speed), to worker zyada pieces assemble karega utne hi time mein.

4.2 Motherboard (Maa Board)

Ye ek badi si printed circuit board hai jiske andar saare important parts ek doosre se judte hain.



- **Analogy:** Ek city ka main highway aur foundation. Saari building (CPU, RAM), park (Graphics Card), transport (data) isi platform par baithte hain.
- **Chipset:** Motherboard ka traffic controller. CPU, RAM, storage, USB sab ke beech data flow isi se manage hota hai.

4.3 Memory Hierarchy (Yaaddaasht Ka Star)

CPU ko kaam karne ke liye data chahiye. Data storage ke different level hain, speed aur price ke hisaab se.



CPU sabse pehle Registers mein dekhta hai, fir Cache, fir RAM, aur fir HDD/SSD. Jaise-jaise neeche jaate hain, speed aur cost kam hoti hai, lekin storage size badhta hai.

4.4 RAM vs ROM (Donon ki Kahani)

Yeh comparison table hamesha yaad rakhiye — exam aur interview ki jaan hai.

Feature	RAM (Random Access Memory)	ROM (Read Only Memory)
Full Form	Random Access Memory	Read Only Memory
Volatility	Volatile (Temporary). Power off, data gayi!	Non-Volatile (Permanent). Data power off hone par bhi safe.
Read/Write	Read and Write dono.	Mostly Read-only. Likhna mushkil ya special process.
Use	Running programs, current tasks, game chalana.	Firmware store karna, jaise BIOS/UEFI.
Speed	Bahut tez.	RAM se slow.
Analogy	Study Table. Padhai kar rahe ho, kitabein khuli hain. Table fold kiya to sab band.	Bookshelf. Kitabein permanently shelf mein hain, utha kar padh lete hain.
Types	SRAM, DRAM (DDR3, DDR4,	PROM, EPROM, EEPROM,

	DDR5)	Flash.
--	-------	--------

4.5 Types of ROM (ROM Ke Avatar)

ROM bhi ek hi type ka nahi hota, technology ke saath evolve hua hai.

- **PROM (Programmable ROM):** Ek baar kharida, ek baar program kiya (jaise CD-R).
- **EPROM (Erasable Programmable ROM):** Isse UV (Ultraviolet) light se mitaya jaa sakta tha. (Bohot jhanjhat tha).
- **EEPROM (Electrically Erasable Programmable ROM):** Isse electricity se program aur erase karo, baar-baar. BIOS isi ka example tha.
- **Flash Memory:** Yehi to aaj ka hero hai. EEPROM ki advanced type, bahut fast aur durable. **SSD, USB drives, Memory cards** sab Flash memory hain. BIOS ki jagah ab **UEFI** ne le li, jo Flash memory mein store hota hai.

4.6 BIOS vs UEFI (Firmware Ki Jung)

Firmware woh software hai jo hardware ko directly control karta hai, permanently chip mein stored rehta hai.

- **BIOS (Basic Input/Output System):** Purana system. Computer ON hote hi sabse pehle yehi jaagta hai. Hardware check karta hai (POST - Power On Self Test), aur phir OS load karta hai. Iska interface simple text-based hota hai, mouse use nahi kar sakte. Sirf 2TB tak ki drive se boot kar sakta hai.
- **UEFI (Unified Extensible Firmware Interface):** BIOS ka modern aur powerful avatar. Graphical interface, mouse support, zyada secure (Secure Boot), aur 9 Zettabyte tak ki drives ko support karta hai. **BIOS is old, UEFI is new. UEFI is mini OS.**

4.7 Important Internal Components

- **CMOS:** Ek chhoti si chip. BIOS/UEFI ki settings store karti hai (date, time, boot order). Iske liye motherboard par ek chhoti si battery lagi hoti hai (Coin Cell).
 - **SMPS (Switched-Mode Power Supply):** Power supply unit. Wall socket se AC current leta hai, computer ke components ke liye smooth DC current mein badalta hai.
 - **Heat Sink & Fan:** CPU aur GPU bahut garam hote hain. Heat sink ek dhatu (aluminium/copper) ka piece hai jo garami ko absorb karta hai. Fan us garami ko bahar nikaal deta hai. Inke beech mein thermal paste lagaya jata hai for better contact.
-

✔ Topic 4: Summary & Revision

Component	Simple Role
CPU	Dimaag (Brain). ALU (Calculator), CU (Manager), Registers (Tiny notepad).
Motherboard	Foundation highway jahan sab kuch judta hai.
RAM	Volatile study table. Fast, temporary.
ROM	Non-volatile bookshelf. Slow, permanent firmware.
BIOS/UEFI	Sabse pehle jaagne wala firmware. UEFI is newer, better.
Clock Speed	Processor ki heartbeat. GHz mein napne.
Heat Sink	CPU ki AC/cooler.

? MCQs

1. **CPU ke kis part ko 'Manager' kehte hain?**

- A) ALU
- B) Register
- C) Control Unit (CU)
- D) Cache

2. **Nimnalikhit mein se kaunsi memory Volatile hai?**

- A) SSD
- B) RAM
- C) ROM
- D) USB Drive

3. **BIOS ka modern replacement kya hai?**

- A) CMOS
- B) UEFI
- C) SMPS
- D) PROM

4. **CPU ki speed napne ki unit kya hai?**

- A) Watt
- B) Volt

- C) Gigahertz (GHz)
- D) Kilometer

(Answers: 1-C, 2-B, 3-B, 4-C)

Interview Questions

1. **Q:** Difference between RAM and ROM?
 - **A:** (Table mein se 3-4 points bol dein, jaise) RAM is Volatile, fast, used for running programs, like a study table. ROM is Non-Volatile, stores permanent firmware, like a bookshelf.
 2. **Q:** What is the role of the Control Unit?
 - **A:** The Control Unit acts like a traffic policeman or an orchestra conductor inside the CPU. It doesn't process data itself but directs and coordinates all operations by fetching, decoding, and executing instructions, managing the flow of data between CPU and other components.
-

Practice Questions

1. Apne mobile phone ki memory ko Memory Hierarchy (Cache, RAM, Storage) se compare kariye. Example dijiye: jab aap PUBG khelte hain, to loading kyun lagti hai?
2. Ek scenario sochiye: Aap MS Word mein ek document type kar rahe hain aur achanak bijli chali jaati hai. Aapka unsaved data kho jaata hai. Isme RAM aur HDD ki kaunsi quality responsible hai?
3. BIOS aur UEFI ke beech ka difference apne shabdon mein likhiye. (Hint: Interface aur Security ke baare mein sochiye).